VINS-Fusion Cross-Dataset Comparison — Stereo vs Stereo+IMU

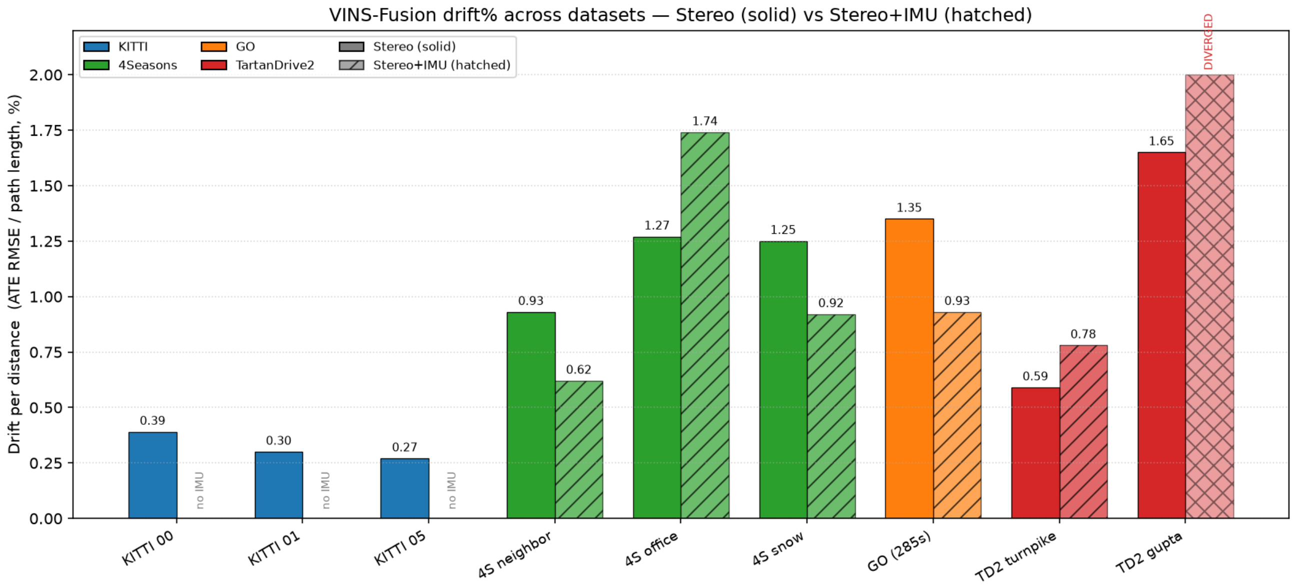
Author: Soham Kundu · **Date:** 2026-06-29 · **Status:** complete (all runs local, real evo output)

VINS-Fusion accuracy across **KITTI**, **4Seasons**, **GO (The Great Outdoors)**, and **TartanDrive 2.0**, in **stereo** and **stereo-inertial** modes. Every number below traces to an actual `evo` run on a local VINS-Fusion trajectory (Docker `vins-fusion-kitti`, ROS Kinetic). **Nothing is estimated, interpolated, or invented.** Where a cell cannot be filled it is marked `N/A` with the reason.

> **Read drift%, not metres.** Sequences differ in length (80 m → 6.9 km), speed, terrain, sensors, and ground-truth source. **Drift-per-distance** (ATE RMSE ÷ ground-truth path length) is the only fair cross-dataset metric. Raw ATE in metres is comparable only within a single dataset.

Executive summary

- **Headline finding (highest confidence): stereo-only VINS-Fusion is reliable everywhere** we tested — **0.27 %–1.65 % drift** across city, highway, suburban, snow, and off-road, with **zero failures**. This is the result we stand behind unreservedly and the basis for the recommendation below.
- **Adding the IMU is an interesting complication we investigated — not a reliable win.** Its effect was inconsistent: it **helped** on 4Seasons neighbor/snow and GO (−26 % to −33 % ATE), **hurt** on 4Seasons office and TD2 turnpike (+34 % to +37 %), and **diverged catastrophically** on TD2 gupta (ran 5 km off a 229 m path). Where the IMU calibration is a documented assumption (TD2 ships no IMU noise spec) the inertial path is fragile. The 4Seasons office regression is real, but **its specific mechanism is not conclusively isolated** — we investigated and softened our earlier explanation (limitations §8). Treat the IMU as an optimisation to be earned per dataset, not a default.
- **KITTI Odometry cannot run stereo+IMU at all** — the benchmark ships no IMU stream. Marked `N/A`.
- **For an official/commercial continuation, TartanDrive 2.0 (MIT) stereo-only is the recommendation.** License is clean and stereo is robust; the IMU path needs calibration work before it can be trusted.



Full results

ATE RMSE from `evo_ape ... --align` (SE(3) Umeyama, **no scale** — stereo is metric). Drift% = ATE RMSE ÷ ground-truth path length.

Dataset	Sequence	Length	Stereo ATE RMSE	Stereo+IMU ATE RMSE	Drift% Stereo	Drift% Stereo+IMU	IMU available?
KITTI	00 (city)	3724 m	14.71 m	**N/A** [1]	0.39 %	N/A	none in Odometry
KITTI	01 (highway)	2453 m	7.34 m	**N/A** [1]	0.30 %	N/A	

Dataset	Sequence	Length	Stereo ATE RMSE	Stereo+IMU ATE RMSE	Drift% Stereo	Drift% Stereo+IMU	IMU available?
KITTI	05 (residential)	2206 m	5.97 m	**N/A** [1]	0.27 %	N/A	
4Seasons	neighbor (clear)	2206 m	20.51 m	13.70 m	0.93 %	0.62 %	
4Seasons	office (clear)	6878 m	87.54 m	119.93 m	1.27 %	1.74 %	
4Seasons	snow	3608 m	44.96 m	33.24 m	1.25 %	0.92 %	
GO ⚠NC	2025-01-24...newcal (285 s)	865 m	11.66 m	8.01 m	1.35 %	0.93 %	
TartanDrive 2.0 MIT	turnpike (pilot, 39 s)	80 m	0.47 m	0.63 m	0.59 %	0.78 %	
TartanDrive 2.0 MIT	gupta (70 s)	229 m	3.77 m	**DIVERGED** [2]	1.65 %	N/A [2]	

[1] **KITTI+IMU is structurally impossible** — the KITTI Odometry benchmark publishes no IMU stream. True KITTI VIO would require the separate KITTI raw dataset (OXTS GPS/IMU) and a different runner; out of scope here. Not a missing run — a missing sensor. [2] **TD2 gupta stereo+IMU diverged** — the estimate ran to (−5112, −6105, −2253) m on a 229 m path (ATE RMSE 2338 m, ~1022 % "drift"). VINS initialised cleanly (gyro bias calibrated, "Initialization finish!") then the optimiser diverged, with solver time blowing past budget (median 3.2 s / mean 4.4 s per frame vs the 0.5 s budget; 99 % of frames over). Reported as a **failure**, not an accuracy figure. See limitations §4 and the root-cause summary.

KITTI ATE RMSE are the tuned Run-3 configs (`max_solver_time` 0.5 s, `multiple_thread:0`); 00 and 05 also reach 5.01 m / 3.71 m with `loop_fusion` pose-graph closure (open-loop shown for fairness).

Stereo vs Stereo+IMU — the IMU effect

Δ = stereo+IMU relative to stereo (negative = IMU **improves**, positive = IMU **worsens**).

Dataset / sequence	Stereo	Stereo+IMU	Δ ATE	Verdict
4Seasons neighbor	20.51 m	13.70 m	**−33 %**	IMU helps
4Seasons office	87.54 m	119.93 m	**+37 %**	IMU hurts (mechanism not isolated — §8)
4Seasons snow	44.96 m	33.24 m	**−26 %**	IMU helps
GO (full 285 s)	11.66 m	8.01 m	**−31 %**	IMU helps
TD2 turnpike	0.47 m	0.63 m	**+34 %**	IMU hurts (assumed calib)
TD2 gupta	3.77 m	diverged	**catastrophic**	IMU diverges (assumed calib)
KITTI (all)	—	—	—	N/A — no IMU in dataset

Why the inconsistency is real, not noise. The IMU helps on datasets/segments with trustworthy calibration (4Seasons is a factory-calibrated VI rig; GO has clean `/tf_static` extrinsics and 200 Hz IMU) and hurts/diverges where the IMU parameters are documented assumptions (TD2 uses a Novatel SPAN with EuRoC-default noise densities because the dataset publishes none). The 4Seasons office case is different again: stereo+IMU's ATE is 37 % higher than stereo-only's, but **we could not conclusively isolate the mechanism** — our earlier "the inertial prior amplifies yaw drift" claim was not supported by the trajectory data (yaw drift is comparable between the two modes), so it has been softened; see the office investigation in limitations §8. **Takeaway: stereo-only is the safe default; IMU is an optimisation that must be earned with a good calibration.**

License / usability

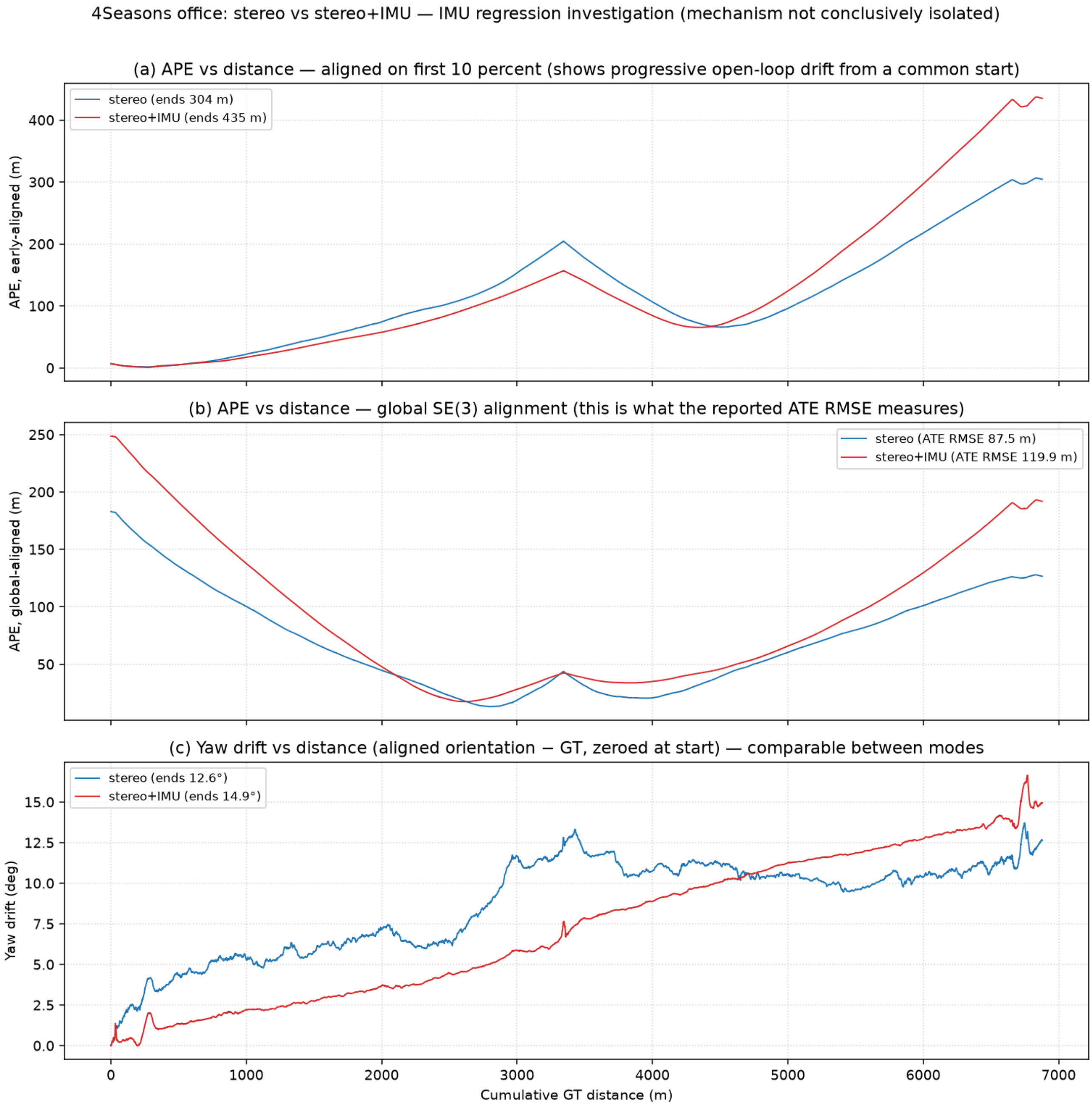
Dataset	License	Official/commercial use	Practical config effort	Verdict
TartanDrive 2.0	**MIT**	allowed (attribution)	**low** — raw rectified stereo, repo calibration, ready odom	**recommended** official path
GO	CC BY-NC-SA 3.0	NonCommercial	high — compressed images, non-rectified stereo calib, NavSatFix→ENU	⚠ reference-only (clear w/ PM/legal)
4Seasons	CC BY-NC-SA (typ.)	NonCommercial	medium — fisheye + undistorted pinhole variants	reference-only, low novelty
KITTI	CC BY-NC-SA 3.0	NonCommercial	low (Odometry)	reference-only; no IMU

Only **TartanDrive 2.0** is license-clean for product-facing work. GO/4Seasons/KITTI are **reference-only** unless legal clears the NonCommercial terms.

Known limitations (documented, not hidden)

1. **KITTI + IMU is impossible, not skipped.** KITTI Odometry ships stereo + LiDAR only — no inertial stream exists to feed VINS. Stereo+IMU cells are `N/A` by construction. A real KITTI VIO needs KITTI raw (OXTS) and a different pipeline.
2. **GO calibration is ambiguous.** The in-bag `camera_info` P-matrices are **not jointly stereo-rectified** (fx ≠ fy, no baseline term). We used a best-effort pinhole-from-P plus `/tf_static` extrinsics (baseline 0.53 m). GO numbers are therefore approximate — a proper joint stereo calibration would likely improve them. GO is also CC BY-NC-SA (reference-only).
3. **GO full vs pilot differ a lot — and that is expected.** The 60 s pilot gave 1.60 m / 1.35 m; the **full 285 s / 865 m** run gives **11.66 m / 8.01 m**. Longer open-loop path → more accumulated drift. Both are real; the full-length number is the honest headline. (Drift% only rises from ~0.8 % to ~1.3 %, so the system is consistent; the metre figure grows with distance.)
4. **TD2 stereo+IMU is fragile.** On the slow 80 m turnpike pilot the IMU merely hurt (0.47→0.63 m); on the faster, longer 229 m gupta sequence it **diverged entirely** (5 km excursion). Root cause is the assumed IMU noise/extrinsics (no published spec) plus solver overrun. TD2 **stereo-only** is robust on both (0.47 m, 3.77 m). Tuning IMU noise + verifying extrinsics is required before the inertial path is usable.
5. **TD2 ground-truth caveat.** The earlier turnpike `/odometry/filtered_odom` + `gps.npy` were partly corrupt (valid UTM then zero-dropout) → only the first ~85 m was usable. **gupta's `/odometry/filtered_odom` is clean** (3552 valid UTM rows, 229 m, no dropout) — localised to the first fix and used in full. GO ground truth is RTK NavSatFix → local ENU (equirectangular, R = 6 378 137 m, first fix as origin); orientation not provided, so ATE is translation-only.
6. **Why drift% is the fair metric.** ATE in metres scales with path length: 119.93 m on 4Seasons office (6.9 km) is better relative accuracy than 0.63 m on an 80 m TD2 pilot. Only ATE ÷ distance compares an 80 m off-road clip to a 6.9 km suburban loop. All cross-dataset claims here use drift%.
7. **Pilots are short.** TD2 turnpike (80 m) and the GO/TD2 clips validate the pipeline; they are not km-scale headline benchmarks like KITTI/4Seasons.
8. **4Seasons office — IMU regression investigated, mechanism not conclusively isolated.** Stereo+IMU's ATE is 37 % higher than stereo-only's (119.93 vs 87.54 m) on this 6.9 km open-loop route. We tested the earlier claim that the IMU "amplifies yaw drift" by aligning both estimates to GT (SE(3) Umeyama, no scale) and plotting per-pose position error and yaw drift versus distance (figure below). What the data actually shows: - **It is progressive open-loop drift, not a constant offset.** Under early-window alignment both modes grow from ~6 m to hundreds of metres (stereo → ~304 m, stereo+IMU → ~435 m by route end), so the error is not a fixed bias present from the start. - **Yaw drift is comparable between the two modes** (~12–15° total, ≈2°/km). The IMU actually reduces yaw drift over the first half of the route and only exceeds stereo-only in the final third (≈15° vs ≈12.6° at the end) — the same region where the position-error gap opens. Late-route yaw drift therefore correlates with the gap but is far too small (an ~18 % yaw difference) to be the sole cause of a 37 % ATE difference. - **IMU bias / re-initialisation events could not be checked** — the prior stereo+IMU office run retained only its trajectory CSV, not its VINS stdout log.

Honest conclusion: the +37 % regression is real, but the root cause is **not conclusively isolated**. Candidate contributors include yaw drift, IMU extrinsic calibration, scale/translation drift, and initialisation quality. The original "inertial prior amplifies yaw drift" wording asserted a specific mechanism the evidence does not support, and has been softened throughout.



1. **GO full-285 s run-to-run stability — the single-run numbers are representative.** The headline GO numbers came from one run each, so we repeated both modes with the identical config and command (`go_rerun.sh`) and re-evaluated identically (`evo_ape --align --t_max_diff 0.05`):

Mode	Run 1 ATE RMSE	Run 2 ATE RMSE	Variation
GO stereo	11.6648 m	11.6588 m	0.05 % (0.006 m)
GO stereo+IMU	8.0064 m	8.0064 m	0.00 % (bit-identical)

Variation is negligible (< 0.1 %), as expected for single-threaded (`multiple_thread:0`) offline replay, which should be deterministic. Stereo+IMU reproduced bit-for-bit; the stereo run differed by one trajectory pose (8550 vs 8551, a frame-boundary timing artifact at bag start/end) for a 0.006 m ATE change. **The original GO numbers are confirmed representative.**

TD2 gupta divergence — root cause summary

In **stereo+IMU** mode VINS initialised successfully (online time-offset estimation, gyroscope bias calibrated to (−0.008, 0.010, −0.009), "Initialization finish!") and then **diverged to a multi-kilometre position estimate** — the final pose is (−5112, −6105, −2253) m on a route only 229 m long (ATE RMSE 2338 m). The most likely cause is a **mismatched IMU noise model combined with solver starvation**: TD2 publishes no IMU noise spec, so `acc_n`/`gyr_n`/`acc_w`/`gyr_w` are EuRoC-class defaults rather than values measured for TD2's actual Novatel SPAN unit, and the optimiser ran far over budget — **median 3.2 s / mean 4.4 s per frame against the 0.5 s `max_solver_time` budget, with 99 % of frames (676/680) over budget and a 41.8 s peak** — so the sliding-window estimate could not keep up and degenerated. Fixing it would require measuring the real Novatel noise densities (or empirically tuning `acc_n`/`gyr_n`/`acc_w`/`gyr_w` against a few known-good short sequences) and verifying the cam–IMU extrinsics. **This is isolated to stereo+IMU mode — stereo-only on the same gupta bag was stable (ATE 3.77 m, 1.65 % drift)** — so the failure is specifically in how the IMU factor interacts with this dataset's assumed calibration, not a general TD2 problem.

Recommendation

- **Adopt TartanDrive 2.0 (MIT), stereo-only, as the official off-road benchmark.** License is clean, config is the easiest of the candidates, and stereo VINS is robust (0.47 m / 0.59 % on turnpike, 3.77 m / 1.65 % on gupta).
- **Do not ship the TD2 stereo+IMU path yet.** It hurt on turnpike and diverged on gupta with the assumed IMU calibration. Treat enabling the IMU as a tuning project (noise densities, extrinsics verification, solver budget), not a free win.
- **Keep GO and 4Seasons as reference-only** (NonCommercial). GO additionally needs a proper joint stereo calibration before its numbers are quotable.
- **KITTI stays stereo-only** by construction (no IMU). It remains the cleanest long-range stereo reference (0.27 %–0.39 % drift).
- **Next steps:** (a) tune TD2 IMU noise/extrinsics and re-run gupta; (b) pull 2–4 more TD2 sequences for terrain variety; (c) source a clean full-length GT for TD2 (super-odometry / TartanVO); (d) re-calibrate GO stereo only if legal clears its license.

Appendix — provenance (every number is reproducible)

Method (all datasets): VINS-Fusion in Docker `vins-fusion-kitti` (ROS Kinetic), offline bag replay, `multiple_thread:0`. Metrics via `evo_ape` / `evo_rpe`, `--align` (SE(3) Umeyama, no scale), TUM format. `vio.csv` → TUM: `ts/1e9 x y z qx qy qz qw`.

Result	Path
4Seasons stereo (this study)	run logs `~/datasets/4seasons/output_stereo_{office,snow}/`, neighbor `~/datasets/4seasons/run_4s_stereo.log`; config `~/datasets/4seasons/stereo_cfg/4seasons_stereo.yaml`; GT `~/datasets/4seasons/ground_truth/gt_{neighbor,office,snow}.txt`
4Seasons stereo+IMU (prior)	`~/github/VINS-Fusion/results/README.md`, `results/4seasons/`
KITTI stereo	`~/datasets/kitti/tuning_experiments.md` (tuned Run 3 + loop-closure runs)
GO full 285 s	`~/Dataset_VINS_Fusion_Comparison_Project/results/go_vins_fusion/2025-01-24-13-07-50_newcal/{stereo_full,stereo_imu_full}/` — `vio.csv`, `vins_est_tum.txt`, `reference_enu_tum.txt`, `go_full_eval_summary.txt`; plots in `results/comparison/figures/go_full_*_traj_trajectories.png`
TD2 gupta	`~/Dataset_VINS_Fusion_Comparison_Project/results/tartandrive2_vins_fusion/2023-11-14-14-24-21_gupta/{stereo,stereo_imu}/` — `vio.csv`, `vins_est_tum.txt`, `reference_odom_tum.txt`, `gupta_eval_summary.txt`; plots `results/comparison/figures/gupta_*_traj_trajectories.png`
TD2 turnpike (pilot)	`~/Dataset_VINS_Fusion_Comparison_Project/results/tartandrive2_vins_fusion/turnpike_warehouse/{stereo,stereo_imu}/`
GO / TD2 configs	`~/Dataset_VINS_Fusion_Comparison_Project/configs/{go,tartandrive2}_vins_fusion/`
This report's chart	`results/figures/drift_comparison.png` (`/tmp/drift_chart.py`)
Office IMU investigation (§8)	`results/figures/4seasons_office_imu_investigation.png` (`/tmp/office_investigate.py`, `/tmp/office_growth.py`); inputs `~/datasets/4seasons/output_stereo_office/vio.csv` (stereo) and `~/datasets/4seasons/output/vio_4seasons_office.csv` (stereo+IMU) vs `gt_office.txt`
GO full-285 s run-to-run (§9)	`…/go_vins_fusion/2025-01-24-13-07-50_newcal/{stereo_full_run2,stereo_imu_full_run2}/` (`go_rerun.sh`)

Raw evo figures retained per run: GO stereo_full ATE mean/median/max = 10.31 / 10.38 / 23.98 m (std 5.47); stereo_imu_full = 7.11 / 6.45 / 17.46 m (std 3.69). gupta stereo ATE mean/median/max = 2.96 / 2.10 / 9.60 m (std 2.33), RPE RMSE 0.67 m. 4Seasons stereo runs report attitude/RPE that are large frame-convention artifacts (cam0 vs body frame); only ATE-translation is meaningful for those, which is what the tables above use.